

Learning Contact-Aware World Action Models via Force Prediction for Robot Manipulation

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Abstract—World action models (WAMs) couple action generation with visual future prediction to improve manipulation, but visual information alone provides limited supervision for the subtle contact dynamics that determine task success. Consequently, learned world representations may emphasize appearance and coarse motion rather than contact dynamics. This paper presents CA-WAM, a contact-aware WAM that uses future force prediction and geometric alignment to learn world representations without requiring force measurements during inference. In particular, the normalized 3D end-effector force tokens are appended to the world expert within the WAM framework. Subsequently, video, force, and action flows are jointly optimized via conditional flow matching. An asymmetric attention mask enables the model to learn contact-aware world representations while maintaining efficient action-only inference. Training-only geometric alignment further regularizes intermediate world representations with multi-view geometry. CA-WAM is comprehensively evaluated on LIBERO and Meta-World, with average success rates of 98.5% and 94.9% respectively. It is further validated on five challenging real-world tasks spanning general and contact-rich manipulation with an average success rate of 71.0%, outperforming Fast-WAM by 36.5%. Ablations confirm that policy improvements are driven by force supervision and geometric alignment, highlighting future force as an effective training-time physical prior for world action models.

I. INTRODUCTION

Vision-language-action (VLA) models have advanced robot manipulation by learning policies conditioned on visual observations, language instructions and proprioceptive states, often leveraging pretrained vision-language representations [1]–[3]. While these policies can implicitly capture interaction patterns from demonstrations, their action objectives do not necessarily provide explicit supervision for predicting how the environment evolves. A complementary line of research explores WAMs, which couple action learning with predictive world modeling to learn from future visual observations [4], [5]. Recent efficient WAMs retain predictive training while removing future video generation at inference, suggesting that its benefits can be retained in the representations used for action generation [6].

Despite these advances, visual prediction alone provides limited supervision for contact dynamics. Many manipulation failures happen at the moment where observations provide insufficient information. As shown in Fig. 1, before motion becomes visibly distinguishable, similar RGB observations may correspond to free motion, impending contact, stable support, slip, or obstruction. These states call for different actions, but visual observations provide no direct supervision for separating them. Therefore, the key limitation lies not in insufficient visual fidelity, but in the absence of information

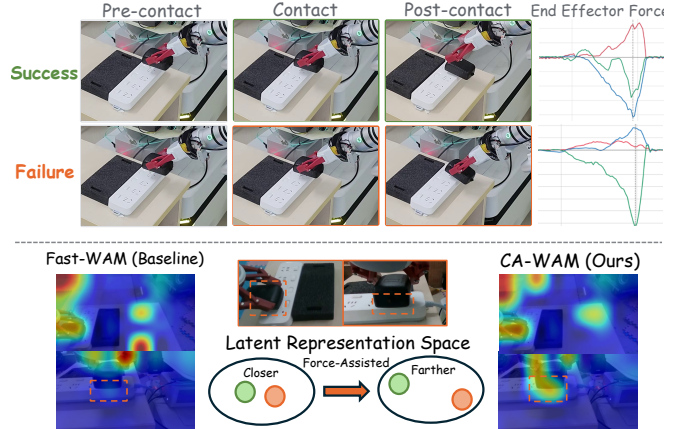


Fig. 1: Motivation for contact-aware world modeling. Top: Successful and failed executions exhibit similar RGB observations but different force trajectories. Bottom: Future force supervision directs action-to-visual attention toward contact-relevant regions (outlined in red) and helps separate representations of visually similar states in latent space.

that directly reflects the underlying physical interaction.

One way to mitigate the ambiguity of visually similar contact states is to incorporate force or tactile information into WAMs. End-effector force measurements provide a compact signal of physical interaction and can serve as supervision for contact dynamics. Existing methods exploit such information by incorporating measured force into policy inputs, distilling force features, or jointly predicting actions and contact signals [7]–[10]. However, these approaches primarily integrate physical information into action generation. Using force supervision to improve world representations while retaining an efficient action-only inference path remains less explored. This raises a key question: can future force prediction serve as training-time supervision for learning contact-aware world representations, without requiring measured force or force prediction at deployment?

This paper proposes a Contact-Aware World Action Model (CA-WAM), which learns contact-aware world representations through future 3D force supervision using a mixture-of-transformers (MoT) architecture with an asymmetric attention mask. During joint conditional flow matching training, future video tokens attend to force tokens, while action queries access only current-frame visual and action tokens. This design enables force-assisted representation learning during training while retaining efficient action-only inference without measured force or future video and force generation. Training-only VGGT [11] alignment enriches visual

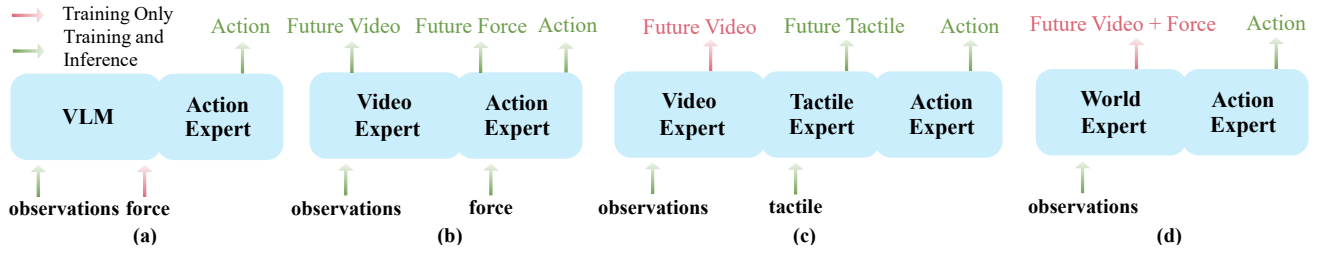


Fig. 2: Overview of force integration in manipulation policies. (a) FD-VLA [8] uses force distillation to improve policy without measured force at inference. (b) FAWAM [9] uses action-side force prediction to correct the policy. (c) VT-WAM [10] uses the MoT architecture to introduce a tactile expert. (d) Our CA-WAM combines future force and video prediction during training to learn contact-aware world representations, without requiring measured force or future video generation at inference.

representations with multi-view geometric priors to support force modeling and action learning. Optional contact-gated replanning uses predicted force trajectories to shorten the action execution horizon [12] near anticipated contact.

Our contributions are threefold:

- We propose CA-WAM that integrates future 3D end-effector force prediction into the world expert to learn contact-aware world representations, while an asymmetric attention mask enables efficient inference.
- A training-only geometric alignment method is proposed to regularize world representations with multi-view geometric priors for force and action prediction.
- A contact-aware dynamic replanning strategy is developed to adjust the action execution horizon based on predicted force.

II. RELATED WORK

A. Vision-Based World Action Models

Generalist VLA policies, such as OpenVLA, π_0 , and $\pi_{0.5}$ [1]–[3], map visual-language inputs to robot actions. World action models extend this paradigm through predictive world modeling, following two complementary lines. The *imagination-oriented* line uses predicted visual futures for action generation, as explored by LingBot-VA and Motus [4], [5]. The *representation-oriented* line emphasizes predictive training: Fast-WAM [6] retains video–action co-training while removing future video generation at inference. Both lines primarily model visual appearance evolution, but they do not explicitly constrain the model to predict future contact forces associated with current observations, task and the distribution of demonstrated behaviors. So visually similar free-space and contact states may remain close in the learned representation. This is particularly limiting when the action decoder should attend to a small local interaction region. For example, attention needs to be paid to the interface between the gripper and the object or the object and the surface before significant visual changes occur during contact. Following the representation-oriented line, CA-WAM introduces future force supervision into the world expert to encourage contact-aware representations for manipulation.

B. Force Integration of Manipulation Policies

Force-aware policies incorporate physical information through different mechanisms. ForceVLA [7] fuses measured

wrenches through a force-aware MoE, while FD-VLA [8] distills force information into a token conditioned on vision and robot state for inference without measured force. FAWAM [9] combines action–wrench prediction with real-time force correction, and TacForeSight [13] uses wrist wrenches to guide tactile prediction and policy fusion. VT-WAM [10] jointly predicts video, tactile deformation, and actions through three MoT experts. Despite their benefits for contact-rich manipulation, several methods require measured force at deployment, introducing sensor dependence and potential sensitivity to measurement noise and drift [7], [9], [13]. FD-VLA [8] removes this requirement, but its latent force token does not directly provide a force trajectory in physical units. Expert routing, residual correction, and tactile expert fusion also introduce additional mechanisms into action generation [7], [9], [10]. Such conditional fusion can benefit contact tasks but may perturb visual-action features when force is less relevant, a risk underexplored on general task suites. As illustrated in Fig. 2, CA-WAM instead uses joint video–force prediction within the world expert to shape visual representations without measured force during inference while preserving an action-only inference path. Optional force prediction provides explicit trajectories for contact-aware replanning.

C. Training-Only Physical and Geometric Targets

Training-only auxiliary objectives enrich visual representations without adding inference inputs. Geometry Forcing [14] aligns diffusion features with a geometric encoder, while VGGT [11] learns multi-view geometric features through 3D prediction tasks. For WAMs, GeoSem-WAM [15] adds geometry and semantic prediction branches, AGRA [16] aligns video features with frozen semantic features, and MECo-WAM [17] transfers VGGT-derived geometric relations through a training-only 4D expert. These approaches primarily supervise geometric structure or visual semantics, without explicitly targeting contact dynamics relevant to action generation. Our CA-WAM aligns visual hidden states of the world expert with frozen VGGT features, complementing future force supervision with multi-view geometric priors. The alignment targets visual features shared by force and action prediction. Geometry provides spatial cues about where interactions occur, while force supervision captures how they evolve, together supporting precise world representations.

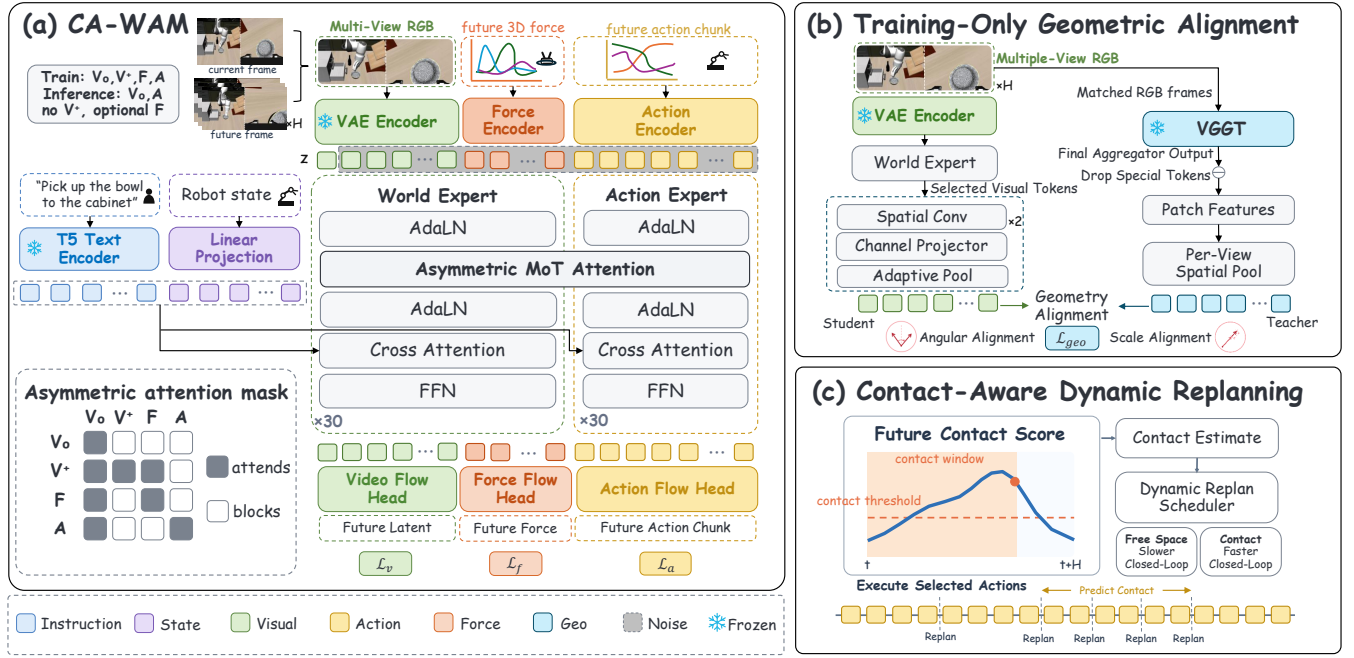


Fig. 3: CA-WAM pipeline. (a) The world expert jointly models future video and force, while the action expert predicts future action under an asymmetric attention mask that lets future visual latents query video and force tokens to model contact-aware world evolution but restricts action queries to the current visual latents and action tokens, enabling force-aware training and future-free efficient action inference. (b) A frozen VGGT teacher provides training-only geometric alignment to intermediate video hidden features in the world expert through a projector. (c) Optional contact-aware replanning uses predicted force changes to select a shorter action execution horizon near anticipated contact and a longer horizon in free space.

III. METHODOLOGY

A. Problem Formulation and Architecture Overview

At decision time t , the robot receives RGB observations $\mathbf{o}_t = \{\mathbf{I}_t^{(v)}\}_{v=1}^{N_c}$ from N_c cameras, a language instruction c , and proprioceptive state $\mathbf{s}_t$. The action expert predicts an H -step action chunk $\mathbf{A}_t = [\mathbf{a}_{t+1}, \dots, \mathbf{a}_{t+H}] \in \mathbb{R}^{H \times D}$, where $\mathbf{a}_t$ is a robot action and D its dimension. During training, each frame additionally provides future frames $\{\mathbf{I}_{t+1:t+H}^{(v)}\}_{v=1}^{N_c}$ and a synchronized end-effector force trajectory $\mathbf{F}_t \in \mathbb{R}^{H \times 3}$ which contains end-effector forces normalized with training set statistics. The multi-view future frames are composed into a video clip and encoded by the Wan variational autoencoder (VAE) [18] into a latent tensor $\mathbf{V}_t \in \mathbb{R}^{C_z \times T_z \times H_z \times W_z}$, where C_z is the channel dimension, T_z is the latent temporal length, and H_z, W_z are the spatial dimensions after VAE downsampling. CA-WAM is trained to model

$$\pi_\theta(\mathbf{A}_t, \mathbf{V}_t, \mathbf{F}_t \mid \mathbf{o}_t, \mathbf{s}_t, c). \quad (1)$$

By default, CA-WAM samples from the action distribution. Optional contact-aware replanning also samples from the force distribution. Neither mode requires measured force or future video generation.

As shown in Fig. 3(a), CA-WAM builds on the two-expert architecture which contains a world expert and an action expert. A camera layout operator $\mathcal{C}$ composes the views of each frame into a canvas. A frozen Wan VAE [18] encodes the resulting training clip into a clean current latent $\mathbf{V}_0$ and future latents $\mathbf{V}_t$. Frozen T5 language features and a learned

projection of $\mathbf{s}_t$ form a shared context sequence $\mathbf{q}_t$, accessed through expert-specific cross-attention.

The world expert is initialized from Wan2.2-TI2V-5B [18] and has 30 transformer blocks with hidden width $d_w = 3072$. The action expert has 30 blocks with width $d_a = 1024$, initialized by scaled interpolation of the world expert. Our method is trained directly on task demonstrations without embodied pretraining. Both experts use 24 attention heads of dimension 128. Their expert-specific projections therefore map different hidden widths to a common attention width of 3072 and back to their respective residual widths.

B. Force-Assisted World Representation Learning

a) *World-Expert force supervision:* For a training window, the action and video targets are perturbed independently by conditional flow matching. For modality $m \in \{v, a, f\}$, define

$$\mathbf{X}_m^{\sigma_m} = (1 - \sigma_m)\mathbf{X}_m + \sigma_m\boldsymbol{\epsilon}_m, \quad \mathbf{u}_m = \boldsymbol{\epsilon}_m - \mathbf{X}_m, \quad (2)$$

where $\mathbf{X}_v = \mathbf{V}_t$, $\mathbf{X}_a = \mathbf{A}_t$, $\mathbf{X}_f = \mathbf{F}_t$, and $\boldsymbol{\epsilon}_m$ is standard Gaussian noise with the same shape as $\mathbf{X}_m$. The scalar $\sigma_m \in [0, 1]$ is the noise level and $\mathbf{u}_m$ is the target velocity. In the implementation, video and action noise levels are sampled independently as $\sigma = 5r/(1 + 4r)$ with $r \sim \mathcal{U}(0, 1)$. Force uses the video noise level, $\sigma_f = \sigma_v$. The observed current latent $\mathbf{V}_0$ remains clean while future video latents $\mathbf{V}_t$ are noised and included in the video flow loss.

After adding video-level noise, the force sequence is

embedded into the world expert width as

$$\mathbf{F} = E_f(\mathbf{F}_t^{\sigma_v}) + \mathbf{P}_f \in \mathbb{R}^{H \times d_w}, \quad (3)$$

where $E_f : \mathbb{R}^3 \rightarrow \mathbb{R}^{d_w}$ is a learned linear projection and $\mathbf{P}_f \in \mathbb{R}^{H \times d_w}$ is a learned temporal embedding. The force tokens share the video noise level embedding with the world stream. Each force token position additionally receives a learned offset for the AdaLN modulation parameters, allowing the model to encode position-dependent force dynamics. Let $\mathbf{V}^+$ and $\mathbf{A}$ denote tokens obtained from $\mathbf{V}_t^{\sigma_v}$ and $\mathbf{A}_t^{\sigma_a}$ respectively. The force tokens $\mathbf{F}$ are appended to the video token sequence $[\mathbf{V}_0; \mathbf{V}^+]$. The world expert jointly processes $[\mathbf{V}_0; \mathbf{V}^+; \mathbf{F}]$, while the action expert processes $\mathbf{A}$. A lightweight flow head maps the final force hidden states to force velocity predictions.

b) Asymmetric MoT attention: The attention mask defines the interactions among current-frame, future video, force, and action tokens. With rows denoting queries and columns denoting keys, the Asymmetric MoT attention mask is

$$\mathcal{M} = \begin{array}{c|cccc} & \mathbf{V}_0 & \mathbf{V}^+ & \mathbf{F} & \mathbf{A} \\ \hline \mathbf{V}_0 & 1 & 0 & 0 & 0 \\ \mathbf{V}^+ & 1 & 1 & 1 & 0 \\ \mathbf{F} & 1 & 0 & 1 & 0 \\ \mathbf{A} & 1 & 0 & 0 & 1 \end{array}, \quad (4)$$

where 1 permits attention between the corresponding groups and 0 blocks it. All groups additionally attend to $\mathbf{q}_t$ through cross-attention.

To be specific, future video queries attend to force tokens, coupling visual prediction to anticipated contact dynamics. Force and action queries attend only to current-frame visual tokens and tokens of their own modality. Thus, neither future video nor force targets enter the action path through forward attention. During backpropagation, gradients from the force prediction loss flow through the current-frame keys and values to update the world expert parameters, thereby shaping the visual representations accessed by the action expert. This provides a route for transferring physical supervision to action generation without requiring force as a policy input.

c) Training objective: Modality-specific flow heads map the final hidden states to velocity predictions $\hat{\mathbf{u}}_v$, $\hat{\mathbf{u}}_a$, and $\hat{\mathbf{u}}_f$. For each modality m , we minimize

$$\mathcal{L}_m = \mathbb{E}[\omega_m(\sigma_m) \|\hat{\mathbf{u}}_m - \mathbf{u}_m\|_{\Omega_m}^2], \quad m \in \{v, a, f\}. \quad (5)$$

$\omega_m(\sigma_m)$ is the noise dependent weight and the set Ω_m contains valid target entries. The expectation is taken over training windows, noise, and noise levels. Before geometric alignment, the training objective is

$$\mathcal{L}_{\text{MoT}} = \lambda_v \mathcal{L}_v + \lambda_a \mathcal{L}_a + \lambda_f \mathcal{L}_f. \quad (6)$$

We use $(\lambda_v, \lambda_a, \lambda_f) = (1, 1, 0.1)$. This joint objective combines visual prediction, action learning, and force supervision to encourage world representations that capture contact dynamics relevant to manipulation.

C. Training-Only Geometric Alignment

As shown in Fig. 3(b), we complement force supervision with a frozen VGGT [11] teacher to regularize the spatial structure of the world representation. We extract patch features from its final aggregator output, excluding camera and register tokens. These features are spatially pooled within each view and arranged according to the camera layout used by $\mathcal{C}$. Teacher frames are selected to align with the video latent sequence. For each matched latent frame i , the resulting teacher target is denoted by $\mathbf{G}_i \in \mathbb{R}^{h_g \times w_g \times d_g}$, where h_g, w_g are its spatial dimensions and $d_g = 2048$.

Let $\mathbf{H}_i^v \in \mathbb{R}^{h_w \times w_w \times d_w}$ denote the visual hidden features of frame i , extracted from the world expert at block index 24 (zero-based) and reshaped into a spatial grid. Here h_w, w_w are the spatial token dimensions and d_w is the world expert hidden width while force tokens are excluded. A trainable projector P_{geo} maps these features to the teacher's spatial and channel dimensions using two 3×3 convolutions with ReLU activations, a 1×1 channel projection, and adaptive average pooling:

$$\hat{\mathbf{G}}_i = P_{\text{geo}}(\mathbf{H}_i^v), \quad \hat{\mathbf{G}}_i \in \mathbb{R}^{h_g \times w_g \times d_g}. \quad (7)$$

Alignment is therefore performed between projected transformer features and teacher features, rather than directly on raw VAE latents.

Let $P = h_g w_g$, and let $\hat{\mathbf{g}}_{ip}, \mathbf{g}_{ip} \in \mathbb{R}^{d_g}$ denote the feature vectors at spatial cell p of $\hat{\mathbf{G}}_i$ and $\mathbf{G}_i$, respectively. The per-frame alignment loss is

$$\ell_i = \frac{1}{P} \sum_{p=1}^P \left[1 - \cos(\hat{\mathbf{g}}_{ip}, \mathbf{g}_{ip}) + \beta \ell_s \left(\log \frac{n(\hat{\mathbf{g}}_{ip})}{n(\mathbf{g}_{ip})} \right) \right], \quad (8)$$

where $\cos$ denotes cosine similarity and $n(\mathbf{g})$ denotes the ℓ_2 norm of feature vector $\mathbf{g}$. The scalar Smooth-L1 loss is defined as $\ell_s(x) = x^2/2$ for $|x| < 1$ and $\ell_s(x) = |x| - 1/2$ otherwise. The cosine term aligns feature directions, while the scale term penalizes differences in log feature norms and we set $\beta = 0.005$.

Over the set $\mathcal{I}$ of current and future latent frames in a minibatch, the geometric objective is

$$\mathcal{L}_{\text{geo}} = \frac{\sum_{i \in \mathcal{I}} w_i \ell_i}{\max\{1, \sum_{i \in \mathcal{I}} w_i\}}, \quad w_i = 1 - \sigma_{v,i}, \quad (9)$$

where $\sigma_{v,i}$ is the sampled video noise level of the clip containing frame i . This clip-level weighting emphasizes cleaner training clips. The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\text{MoT}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}}, \quad (10)$$

where the loss weight $\lambda_{\text{geo}} = 0.1$. Geometric alignment encourages the world expert to encode multi-view spatial structure and object relationships, providing geometry-informed visual representations for force and action prediction. These spatial cues complement force supervision in modeling contact interactions. The teacher and projector are used only during training and are omitted at inference.

D. Contact-Aware Dynamic Replanning

a) *Efficient action inference*: At deployment, the policy uses world representations shaped by training-time force supervision and geometric alignment to generate actions from the current frame, language, and proprioception. The world expert prefills its key-value cache, and the action expert denoises only the action tokens. Because the asymmetric attention mask isolates this pathway from $\mathbf{V}^+$ and $\mathbf{F}$, removing these tokens preserves the training-time information pattern. The policy thus retains the learned contact-aware representations without requiring measured force, the VGGT teacher, future video rollout, or force denoising. The policy generates an action chunk directly from the current observations:

$$\hat{\mathbf{A}}_t \sim \pi_\theta(\mathbf{A}_t \mid \mathbf{o}_t, \mathbf{s}_t, c). \quad (11)$$

b) *Optional contact-aware replanning*: For contact-rich tasks, CA-WAM optionally uses predicted force to adapt the action execution horizon m_t . In addition to generating an action chunk through the standard path, it samples a future force trajectory from the current observation context $(\mathbf{o}_t, c, \mathbf{s}_t)$. The force sampler conditions only on the learned current-frame representation produced by the world expert. Its prediction determines how many actions are executed before replanning.

Let $\hat{\mathbf{f}}_{t+j} \in \mathbb{R}^3$ denote the normalized force prediction at action offset j . We compute a force change score over a look-ahead window $\mathcal{W} \subseteq \{1, \dots, H\}$:

$$r_t = \max_{\tau \in \mathcal{W}} \left\| \hat{\mathbf{f}}_{t+\tau} - \frac{1}{R} \sum_{j=1}^R \hat{\mathbf{f}}_{t+j} \right\|_2, \quad (12)$$

where $1 \leq R \leq H$ is the reference window length. This score measures predicted force deviations from the initial reference level and serves as a proxy for contact transitions.

We maintain a binary scheduling flag χ , initialized to zero. At each decision, hysteresis updates the flag as

$$\chi_t = \begin{cases} \mathbb{I}[r_t > \gamma], & \chi_{\text{prev}} = 0, \\ \mathbb{I}[r_t > \eta\gamma], & \chi_{\text{prev}} = 1, \end{cases} \quad 0 < \eta < 1, \quad (13)$$

where χ_{prev} is the flag from the previous decision. The lower exit threshold $\eta\gamma$ reduces rapid switching. The action execution horizon is then

$$m_t = \begin{cases} m_{\text{contact}}, & \chi_t = 1, \\ m_{\text{free}}, & \chi_t = 0, \end{cases} \quad 1 \leq m_{\text{contact}} < m_{\text{free}} \leq H. \quad (14)$$

Shorter action execution horizons allow the policy to incorporate a new observation after fewer open-loop actions, while longer action execution horizons reduce replanning calls. Alg. 1 summarizes the deployment procedure. Predicted force may also support anomaly detection during real-world execution as an additional cue for safety monitoring.

IV. EXPERIMENTS

Our experiments address five questions: **Q1**: Does CA-WAM improve manipulation performance in simulation and

Algorithm 1 CA-WAM contact-aware replanning

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1: Initialize action-step index  $t \leftarrow 0$  and flag  $\chi \leftarrow 0$ 
2: while the task is not terminated do
3:   Observe RGB views  $\mathbf{o}_t$ , instruction  $c$ , and state  $\mathbf{s}_t$ 
4:   Sample action chunk  $\hat{\mathbf{A}}_t$  using Eq. (11)
5:   Set  $m_t \leftarrow m_{\text{fixed}}$ 
6:   if contact-aware replanning is enabled then
7:     Sample  $\hat{\mathbf{F}}_t$  conditioned on  $(\mathbf{o}_t, \mathbf{s}_t, c)$ 
8:     Compute  $r_t$  using Eq. (12)
9:     Update  $\chi$  using Eq. (13)
10:    Select  $m_t$  using Eq. (14)
11:   end if
12:   Execute the first  $m_t$  actions
13:   Set  $t \leftarrow t + m_t$ 
14: end while

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on a real robot? **Q2**: Does training-time force supervision improve manipulation performance? **Q3**: Does integrating force prediction into the world expert improve manipulation performance compared with alternative force integration designs? **Q4**: Does geometric alignment improve prediction quality and manipulation performance? **Q5**: Does contact-aware replanning further improve manipulation performance?

A. Simulations

1) *Simulation setup*: We evaluate CA-WAM on the four LIBERO suites [20] and Meta-World MT50 [21], using the difficulty grouping from MWM [22]. We compare against generalist VLAs, including OpenVLA [1], SmolVLA [19], π_0 [2], and $\pi_{0.5}$ [3]; WAM baseline Fast-WAM [6]; and force-aware policies, including FD-VLA [8] and ForceVLA [7]. Fast-WAM serves as the primary baseline for comparison with CA-WAM. We use published LIBERO results for OpenVLA, SmolVLA, π_0 , $\pi_{0.5}$, and Fast-WAM, and published Meta-World results for SmolVLA, π_0 , and $\pi_{0.5}$. All remaining results are from our evaluations, following the experimental settings described in the original papers.

We use LIBERO demonstrations and the Meta-World dataset from SmolVLA [19], replaying trajectories in simulation to collect 3D wrist force measurements. CA-WAM optionally uses contact-aware replanning with task-specific parameters. Otherwise, it uses a fixed action execution horizon. We evaluate CA-WAM and its variants over three training seeds and 50 rollouts per task per seed. Success rates are averaged across seeds and tasks within each suite or difficulty tier.

2) *Simulation Results*: To address **Q1**, we compare CA-WAM with the baselines in Table I. CA-WAM achieves the highest average success rates among the compared methods, reaching 98.5% on LIBERO and 94.9% on Meta-World. These results exceed Fast-WAM by 0.9 and 2.4 percentage points respectively. Improvements span the LIBERO Spatial, Goal, and Long suites and all four Meta-World difficulty tiers, with the largest gain of 4.3 percentage points on Meta-World’s Very Hard tasks.

TABLE I: Success rates (%) on LIBERO and Meta-World. Avg. is the mean over four suites or difficulty tiers, respectively. Emb. PT denotes additional embodied pretraining. Parentheses report changes of percentage points relative to Fast-WAM for CA-WAM, SmolVLA for FD-VLA, and π_0 for ForceVLA.

Method	Emb. PT	LIBERO					Meta-World				
		Spatial	Object	Goal	Long	Avg.	Easy	Medium	Hard	Very Hard	Avg.
OpenVLA [1]	Yes	84.7	88.4	79.2	53.7	76.5	81.7	59.6	66.3	62.8	67.6
SmolVLA [19]	No	90.0	96.0	92.0	71.0	87.3	82.5	41.8	45.0	60.0	57.3
π_0 [2]	Yes	96.8	98.8	95.8	85.2	94.1	71.8	48.2	41.7	30.0	47.9
$\pi_{0.5}$ [3]	Yes	98.8	98.2	98.0	92.4	96.9	86.1	78.2	80.0	82.0	81.6
Fast-WAM [6]	No	98.2	100.0	97.0	95.2	97.6	93.8	92.0	91.8	92.5	92.5
CA-WAM (Ours)	No	99.2	99.8	98.6	96.4	98.5(+0.9)	95.9	92.4	94.3	96.8	94.9(+2.4)
<i>Force variants</i>											
FD-VLA [8] (SmolVLA backbone)	No	72.8	85.6	72.2	40.4	67.8(-19.5)	66.1	60.0	31.7	58.0	54.0(-3.3)
ForceVLA [7] (π_0 backbone)	Yes	66.0	78.4	71.6	41.0	64.3(-29.8)	65.4	46.7	27.5	25.6	41.3(-6.6)
<i>CA-WAM variants</i>											
CA-WAM (Ours)	No	99.2	99.8	98.6	96.4	98.5	95.9	92.4	94.3	96.8	94.9
Action-Force (FAWAM [9] style)	No	97.6	98.8	95.8	92.8	96.2	93.1	91.6	90.2	90.3	91.3
Three-Expert (VT-WAM [10] style)	No	98.2	99.0	96.2	92.2	96.4	93.5	92.3	91.4	92.1	92.3
CA-WAM w/o dynamic replan	No	99.2	99.4	97.4	95.8	98.0	95.5	92.4	94.0	96.5	94.6
CA-WAM w/o force	No	97.2	99.0	96.0	93.4	96.4	94.3	92.2	92.2	93.8	93.1
CA-WAM w/o VGGT	No	97.0	98.8	96.6	92.8	96.3	94.0	92.0	91.9	94.2	93.0
CA-WAM w/o VGGT & force	No	96.6	98.2	95.8	92.6	95.8	93.9	92.1	91.7	92.4	92.5

To address **Q3**, we compare four force integration designs (Table I). FD-VLA and ForceVLA underperform their respective backbones by 3.3 and 6.6 percentage points on Meta-World and 19.5 and 29.8 points on LIBERO, suggesting these force-aware methods do not uniformly improve on their backbones for general tasks. We also compare Action-Force, which predicts force in the action expert following the FAWAM style, and Three-Expert, which uses a dedicated force expert following the VT-WAM style. Their SRs fall below CA-WAM without dynamic replanning by 1.8 and 1.6 points on LIBERO and 3.3 and 2.3 points on Meta-World, respectively, supporting CA-WAM’s force integration design.

3) *Ablation Studies*: To isolate the effects of training-time supervision, all comparisons for **Q2** and **Q4** use a fixed action execution horizon. To address **Q2**, we remove force supervision during training, which reduces SR by 1.6 percentage points on LIBERO and 1.5 points on Meta-World (Table I). These results suggest that future force supervision improves manipulation performance. To address **Q4**, we remove VGGT geometric alignment during training, which reduces SR by 1.7 points on LIBERO and 1.6 points on Meta-World (Table I), indicating that geometric alignment also contributes to manipulation performance. Removing both VGGT alignment and force supervision reduces SR by 2.2 and 2.1 points which supports the benefit of combining force supervision with geometric alignment.

To address **Q5**, we replace contact-aware replanning with a fixed action execution horizon, which reduces SR by 0.5 percentage points on LIBERO and 0.3 points on Meta-World (Table I). This indicates that using predicted force to adapt the action execution horizon provides an additional benefit beyond training-time force supervision.

B. Representation, Prediction and Efficiency Analysis

We first qualitatively compare the spatial attention patterns of Fast-WAM and CA-WAM and then examine the effects of

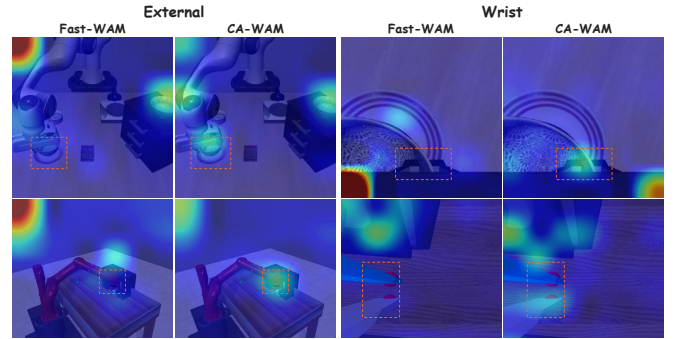


Fig. 4: Qualitative comparison of action-to-visual attention at layer 24 between Fast-WAM and CA-WAM. Warmer colors indicate higher attention weights.

TABLE II: Video generation quality of CA-WAM ablations, averaged across LIBERO and Meta-World.

Variant	VGGT	Force	Latent RMSE↓	PSNR (dB)↑	SSIM↑
CA-WAM	✓	✓	0.542	19.06	0.752
w/o VGGT	×	✓	0.581	18.46	0.735
w/o force	✓	×	0.529	19.44	0.766
w/o VGGT & force	×	×	0.576	18.27	0.740

force supervision and geometric alignment on video generation, extending the analyses of **Q2** and **Q4**. We then evaluate the effect of geometric alignment on force prediction and the trade-off between force prediction accuracy and efficiency.

1) *Qualitative representation analysis*: We compare action-to-visual attention in Fast-WAM and CA-WAM using identical observations and task instructions. In Fig. 4, the first and second rows show examples from LIBERO and Meta-World. CA-WAM exhibits more contact-aware attention around relevant interaction regions. These visualizations suggest that CA-WAM learns world representations that better capture contact-relevant information.

TABLE III: Force prediction accuracy and inference efficiency, averaged across LIBERO and Meta-World.

Predictor	F steps	Latency (ms)↓	MAE (N)↓	ρ_{L2} ↑
CA-WAM	1	246.8	2.062	0.830
CA-WAM	10	404.5	1.986	0.825
CA-WAM w/o VGGT	1	248.1	2.398	0.783
CA-WAM w/o VGGT	10	403.9	2.376	0.791
Action-Force	10	224.6	2.435	0.794
Three-Expert	10	624.7	2.351	0.811

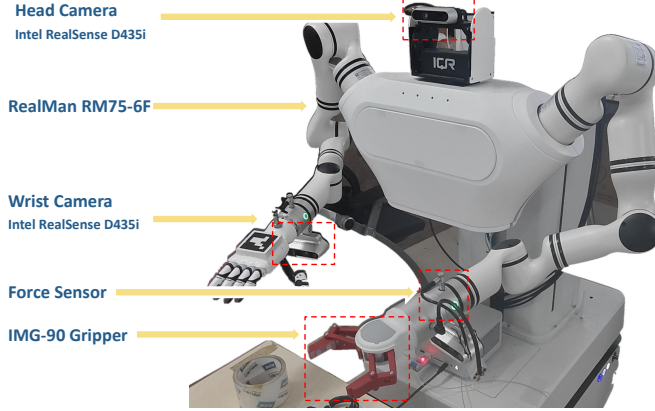


Fig. 5: Real-world experimental setup. We use the left-arm gripper of RealMan RM75-6F platform for all manipulation experiments, with RealSense D435i cameras providing head and wrist views and a wrist force sensor recording forces.

2) *Video prediction*: We evaluate video prediction only in simulation using latent RMSE, PSNR, and SSIM, ablating force supervision, VGGT alignment, or both (Table II). With force supervision enabled, VGGT reduces latent RMSE from 0.581 to 0.542 and increases PSNR from 18.46 to 19.06 dB. However, removing force supervision while retaining VGGT further increases PSNR to 19.44 dB and SSIM to 0.766, despite reducing policy success rates (Table I). These results suggest that force supervision benefits action-relevant world representations rather than visual reconstruction fidelity.

3) *Force prediction and efficiency*: We evaluate force prediction on LIBERO and Meta-World using force-component MAE (N) and force-magnitude Pearson correlation ρ_{L2} (Table III). Latency includes action generation per chunk, and F steps denotes force denoising steps. Without VGGT, 1-step MAE increases by 0.336 N and ρ_{L2} decreases by 0.047, with a similar trend at 10 steps. Increasing sampling from 1 to 10 steps adds 157.7 ms per chunk for only a small MAE improvement and little change in ρ_{L2} , favoring 1-step sampling. CA-WAM achieves lower MAE and higher ρ_{L2} than both Action-Force and Three-Expert while maintaining efficient inference.

C. Real-World Experiments

We complete the evaluation of **Q1** with real-world manipulation experiments.

1) *Experimental setup*: We use the left arm of a RealMan platform, with head and wrist RGB cameras and a wrist

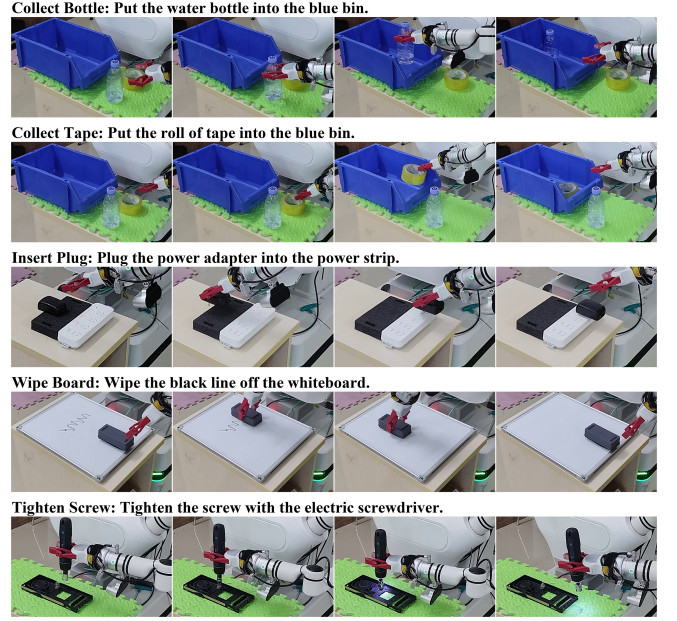


Fig. 6: Real-world manipulation tasks: Collect Bottle, Collect Tape, Insert Plug, Wipe Board, and Tighten Screw. Frames for each task are shown in chronological order from left to right.

TABLE IV: Real-world success counts over 20 trials per task and overall success rates.

Method	General		Contact-Rich			Avg.
	Collect Bottle	Collect Tape	Insert Plug	Wipe Board	Tighten Screw	
$\pi_{0.5}$	17/20	15/20	5/20	10/20	10/20	57.0%
FD-VLA	15/20	12/20	7/20	11/20	9/20	54.0%
ForceVLA	17/20	13/20	7/20	10/20	11/20	58.0%
Fast-WAM	16/20	15/20	4/20	8/20	9/20	52.0%
CA-WAM	18/20	16/20	10/20	13/20	14/20	71.0%

force sensor (Fig. 5). We evaluate two general pick-and-place tasks, Collect Bottle and Collect Tape, and three contact-rich tasks, Insert Plug, Wipe Board, and Tighten Screw (Fig. 6). The contact-rich tasks involve constrained insertion and sustained surface contact. We collect 60 demonstrations per pick-and-place task and 100 per contact-rich task, yielding 420 demonstrations in total. We compare CA-WAM with $\pi_{0.5}$, FD-VLA, ForceVLA, and Fast-WAM, training all methods on the same demonstrations. Each method is evaluated over 20 trials per task under the same initial conditions. $\pi_{0.5}$ and Fast-WAM do not use force information. FD-VLA predicts force features without measured force input, whereas ForceVLA uses measured force [7], [8]. CA-WAM uses predicted force for contact-aware replanning without measured force input.

2) *Evaluation Results*: On the two general pick-and-place tasks, FD-VLA and ForceVLA achieve lower SRs than Fast-WAM and $\pi_{0.5}$ (Table IV), with decreases of 10.0 and 2.5 percentage points compared with Fast-WAM respectively. CA-WAM achieves the highest SR among the evaluated

methods, with improvements of 7.5 points over Fast-WAM and 5.0 points over $\pi_{0.5}$. These results suggest that CA-WAM benefits not only contact-rich manipulation but also general tasks in which force information is less relevant. On contact-rich tasks, FD-VLA and ForceVLA outperform Fast-WAM by 10.0 and 11.7 points, and $\pi_{0.5}$ by 3.3 and 5.0 points respectively. This pattern is consistent with the benefits of force-aware policies for contact-rich manipulation. CA-WAM further improves SR over all compared methods, with gains of 16.7 percentage points over FD-VLA, 15.0 over ForceVLA, 20.0 over $\pi_{0.5}$, and 26.7 over Fast-WAM, suggesting particular benefits for contact-rich manipulation.

Across all five tasks, CA-WAM achieves 71.0% SR, a 36.5% relative improvement over Fast-WAM. These results demonstrate the effectiveness of the complete CA-WAM system across both general and contact-rich manipulation, with larger gains on contact-rich tasks.

V. CONCLUSIONS

We propose CA-WAM, a world action model that learns contact-aware world representations through future force supervision. By jointly predicting future video and 3D end-effector force within the world expert, CA-WAM introduces physical supervision into visual representation learning. Training-only alignment with a frozen VGGT teacher further regularizes these representations with multi-view geometric priors. An asymmetric attention mask allows an efficient action-only inference path. Predicted force remains available during optional contact-aware dynamic replanning. Experiments in simulation and on real robots demonstrate CA-WAM’s effectiveness, while ablation studies validate its architectural design. These findings support future force as a useful training-time physical prior for improving manipulation without requiring measured force at inference. The approach requires synchronized force labels during training, which may be costly to collect at scale. Its contact-aware replanning can adjust frequency but does not regulate interaction forces. Future work will explore scalable force supervision, uncertainty-aware replanning, and integration with low-level force control for broader contact-rich manipulation.

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